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Reconfigurable Radio Systems (RRS);
Radio Reconfiguration related requirements
for Mobile Devices

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Foreword

This European Standard (EN) has been produced by ETSI Technical Committee Reconfigurable Radio Systems (RRS).

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Date of adoption of this EN:	17 May 2018
Date of latest announcement of this EN (doa):	31 August 2018
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Modal verbs terminology

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1 Scope

The scope of the present document is to define the high level system requirements for reconfigurable Mobile Devices enabling the provision of Radio Applications. The work will be based on the Use Cases defined in ETSI TR 103 062 [i.1] and ETSI TR 102 944 [i.2].

2 References

2.1 Normative references

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The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] ETSI TR 103 062: "Reconfigurable Radio Systems (RRS); Use Cases and Scenarios for Software Defined Radio (SDR) Reference Architecture for Mobile Device".

[i.2] ETSI TR 102 944: "Reconfigurable Radio Systems (RRS); Use Cases for Baseband Interfaces for Unified Radio Applications of Mobile Device".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the following terms and definitions apply:

Functional Block (FB): function needed for real-time implementation of Radio Application(s)

NOTE 1: A functional block includes not only the modem functions in Layer1 (L1), Layer2 (L2), and Layer 3 (L3) but also all the control functions that should be processed in real-time for implementing given Radio Application(s).

NOTE 2: Functional blocks are categorized into *standard functional blocks* and *user defined functional blocks*. In more details:

1) *Standard functional blocks* can be shared by many Radio Applications. For example, Forward Error Correction (FEC), Fast Fourier Transform (FFT)/Inverse Fast Fourier Transform (IFFT), (de)interleaver, Turbo coding, Viterbi coding, Multiple Input Multiple Output (MIMO), Beamforming, etc. are the typical category of standard functional block.

2) *User defined functional blocks* include those functional blocks that are dependent upon a specific Radio Application. They are used to support special function(s) required in a specific Radio Application or to support a special algorithm used for performance improvement. In addition, a user defined functional block can be used as a baseband controller functional block which controls the functional blocks operating in baseband processor in real-time and to control some context information processed in real-time.

NOTE 3: Each functional block has its unique name, Input, Output and properties.

network coding: technique in which transmitted data is encoded and decoded to improve network performance

Radio Application (RA): software which enforces the generation of the transmit RF signals or the decoding of the receive RF signals

NOTE 1: The software is executed on a particular radio platform or an RVM as part of the radio platform.

NOTE 2: Radio applications might have different forms of representation. They are represented as:

- source codes including radio library calls of radio library native implementation and Radio HAL calls;
- Intermediate Representations (IRs) including radio library calls of radio library native implementation and radio HAL calls;
- Executable codes for a particular radio platform.

radio library: library of Standard Functional Blocks (SFB) that is provided by a platform vendor in a form of platform-specific executable code

NOTE 1: SFBs implement reference codes of functions which are typical for radio signal processing. They are not atomic and their source codes are typed and visible for Radio Application developers.

NOTE 2: An SFB is implemented through a Radio Hardware Abstraction Layer (HAL) when the SFB is implemented on dedicated HW accelerators. Radio HAL is part of ROS.

Radio Virtual Machine (RVM): abstract machine supporting reactive and concurrent executions

NOTE: A Radio Virtual Machine may be implemented as a controlled execution environment which allows the selection of a trade-off between flexibility of base band code development and required (re-) certification efforts.

reconfigurable mobile device: mobile device with radio communication capabilities providing support for radio reconfiguration

NOTE: Reconfigurable mobile devices include but are not limited to: smartphones, feature phones, tablets, laptops.

resources: hardware resources that a Radio Application needs in active state

NOTE 1: Resources are provided by the reconfigurable Mobile Device (MD), to be used by the Radio Applications when they are active. Radio Applications provide their resource needs (e.g. using operational states) so that the multiradio computer may judge whether these resources are available, in order to ensure non-conflicting operation with other Radio Applications. Resources may or may not be shared in the reconfigurable MD.

NOTE 2: Resources may include processors, accelerators, memory, Radio Frequency circuitry, etc.

3.2 Abbreviations

For the purposes of the present document, the following abbreviations apply:

ASIC	Application Specific Integrated Circuit
BER	Bit Error Rate
CAT	CATegory
CR	Cognitive Radio
FB	Functional Block
FEC	Forward Error Correction
FFT	Fast Fourier Transform
HAL	Hardware Abstraction Layer
IR	Intermediate Representation
LTE	Long Term Evolution
MAC	Media Access Control
MD	Mobile Device
MDRC	Mobile Device Reconfiguration Class
MIMO	Multi-Input Multi-Output
MU-MIMO	Multi User- Multi-Input Multi-Output
PER	Packet Error Rate
PMI	Precoding Matrix Indicator
RA	Radio Application
RAT	Radio Access Technology
RF	Radio Frequency
RI	Rank Indicator
ROS	Radio Operating System
RRS	Reconfigurable Radio Systems
RSSI	Received Signal Strength Indication
RVM	Radio Virtual Machine
Rx	Receive
SDR	Software Defined Radio
SFB	Standard Functional Block
SINR	Signal to Interference-plus-Noise Ratio
SU-MIMO	Single User- Multi-Input Multi-Output
Tx	Transmit
UDFB	User Defined Functional Block
WiFi	Wireless Fidelity

4 Requirement organization and methodology

4.0 General

This clause is containing the description of how the requirements are organized and the related format.

4.1 Requirement organization

As shown in Figure 1, all requirements described in the present document belong to one single category (the functional requirements category). Requirements are, in turn, organized into groups.



Figure 1: Overall requirements structure

4.2 Requirement format

A letter code system is defined which makes a unique identification of each requirement R-<CAT>-<GROUP>-<XX>. Each requirement is constructed as follows:

- R-: Standard requirement prefix
- <CAT>

Code	Category
FUNC	Functional aspects

- <GROUP>: Requirement group identifier. A letter code will be used for this identifier. The three first letters will give the identifier of the group
- <XX>: Requirement identifier within requirement group; range 01 => 99

EXAMPLE: R-FUNC-QOS-01.

4.3 Requirement formulation

A requirement is formulated in such a way that it is uniquely defined. It is built as follows:

Title: <Title Description>

- Description: the description of a requirement will be formulated using one of the following terms:
  - "shall" is used to express mandatory requirements (i.e. provisions that have to be followed)
  - "should" is used to express recommendations (provisions that an implementation is expected to follow unless there is a strong reason for not doing so)
  - "May" is used to express permissible actions (provisions that an implementation is able to follow or not follow)

5 Working assumptions

5.1 Assumptions

5.1.1 Mobile device reconfiguration classes

As it is expected that the reconfiguration capabilities of a Mobile Device will evolve over time, Mobile Device Reconfiguration Classes (MDRC) are introduced. As shown in Figure 2, 7 different classes of reconfigurable MD are introduced (MDRC-0 corresponds to a non-reconfigurable device).

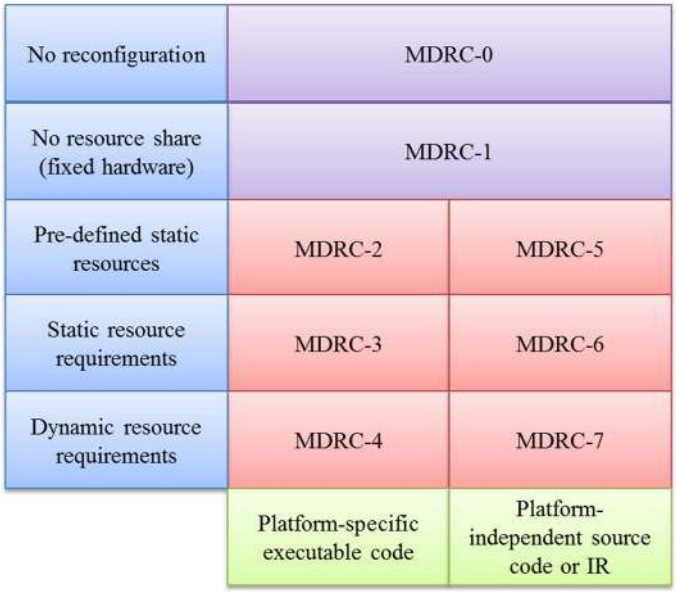


Figure 2: Definition of MDRCs according to reconfiguration capabilities

A reconfigurable MD belongs to a defined class according to the reconfiguration capabilities, which are determined by the type of Resource requirements and the form of the Radio Application Package. Reconfigurable MD classes are defined as follows (see also Figure 2):

1) MDRC-0: No MD reconfiguration is possible

   MDRC-0 represents legacy radio implementations and do not allow for MD reconfiguration (except for bug fixing and release-updates through firmware updates) or exploitation of Cognitive Radio (CR) features. MDRC-0 represents legacy radio implementations and does not allow for MD reconfiguration.

2) MDRC-1: Radio Applications use different fixed Resources

In this scenario, at least some of the radios are implemented with non-software defined radio (SDR) technology, e.g. with dedicated Application Specific Integrated Circuits (ASICs), and are Resource-wise independent of each other. Simple CR functionality may be supported through radio parameter management to the extent which the radio implementations allow.
MDRC-1 implements multiple Radio Applications with fixed Resources allocation and no Resource sharing. The rule for Resource allocation for multiple applications $\{A_1, A_2, \ldots, A_N\}$ can be formulated as follows:

$A_i \rightarrow R_i$, $\forall i \in \{1, ..., N\}$, where R_i denotes Resources allocated for application A_i and $R_i \cap R_j = \varnothing$ for $\forall i \neq j$.
Note that applications can be run concurrently in any combination; a Resource allocation mechanism within separate applications is not specified.

3) MDRC-2: Radio Applications use pre-defined static Resources

MDRC-2 implements multiple Radio Applications but no dynamic Resource management is available. The Radio Applications for MDRC-2 come from a single Radio Application Package which is normally provided by a reconfigurable MD vendor or SDR chipset manufacturer. In this scenario, it is assumed that software radio components in the Radio Application Package are provided in platform-specific executable code.
The rule for the Resource allocation related to multiple applications $\{A_1, A_2, \ldots, A_N\}$ can be formulated as follows: $A_i \rightarrow R_i$, $\forall i \in \{1, ..., N\}$, where R_i denotes Resources allocated for application A_i, if $\exists\ i \neq j$ so that $R_i \cap R_j \neq \varnothing$ then such applications cannot be run concurrently, all other combinations are allowed; a Resource allocation mechanism within separate applications is not specified.

4) MDRC-3: Radio Applications have static Resource requirements

For MDRC-3, a Resource budget is defined for each Radio Application. This budget contains a static Resource measure that represents the worst-case Resource usage of the application, generated at Radio Application compile-time. If an application is being started, the Resource manager installed in a reconfigurable MD of MDRC-3 checks its Resource budget and the sum of all Resource budgets of already running applications, and admits the new application only if the Resources can still be guaranteed for all running applications. In this scenario, it is assumed that software radio components in the Radio Application Package are provided in platform-specific executable code.
The rule for Resource allocation for multiple applications $\{A_1, A_2, \ldots, A_N\}$ can be formulated as follows: $A_i \rightarrow \mathrm{R}(A_i)$, where R denotes total Resources to be shared and $\mathrm{R}(A_i)$ denotes a part of R allocated for A_i; if for $i1, i2, ..., iM \in \{1, ..., N\}$, $M \leq N$, $\mathrm{R}(A_{i1}) \cup \mathrm{R}(A_{i2}) \cup ... \cup \mathrm{R}(A_{iM}) \subset \mathrm{R}$ then applications $A_{i1}, A_{i2}, \ldots, A_{iM}$ can be run concurrently; a Resource allocation mechanism within separate applications is not specified.

5) MDRC-4: Radio Applications have dynamic Resource requirements

This scenario assumes a similar Resource manager in a reconfigurable MD as for MDRC-3, but in addition the Radio Applications have now varying Resource demands based on their current type of activity. Applications have separate operational states for different types of activity, and a Resource budget is assigned to each operational state. In this scenario, it is assumed that software radio components in the Radio Application Package are provided in platform-specific executable code.
Resource management for MDRC-4 can be formulated as follows. Multiple applications $\{A_1, A_2, \ldots, A_N\}$ can be run and each application A_i is divided into tasks $\{t_1(A_i), t_2(A_i), ..., t_k(A_i)\}$. Resource allocation is provided by the Resource manager in a reconfigurable MD for each task $t_j(A_i) \rightarrow \mathrm{R}(t_j(A_i))$.
The rule for task running is exactly the same as for MDRC-3 except that each application should be replaced by a corresponding task. Therefore, if for $i1, i2, ..., iM \in \{1, ..., N\}$, $M \leq N$, $\mathrm{R}(t_{j1}(A_{i1})) \cup \mathrm{R}(t_{j2}(A_{i2})) \cup ... \cup \mathrm{R}(t_{jL}(A_{iM})) \subset \mathrm{R}$ then tasks $t_{j1}(A_{i1}), t_{j2}(A_{i2}), ..., t_{jL}(A_{iM})$ can be run concurrently; a Resource allocation mechanism within separate tasks is not specified.

6) MDRC-5: Radio Applications use pre-defined static Resources, on-device compilation of Software Radio Components

This class corresponds to MDRC-2 with the difference that all or part of the software radio components are provided in the Radio Application Package as platform-independent source code or platform-independent Intermediate Representation (IR), which is compiled on the reconfigurable MD itself. It particularly means that the reconfigurable MD should include a proper compiler in order to convert the source code or IR of the software radio components into an executable code that runs on a given modem chip of a reconfigurable MD. It is assumed that the methods of radio programming and the tools to support this category have become sufficiently standardized so that third-party vendors may create Radio Applications and activate them to different platforms with relative ease. The formal description of the Resource management is the same as for MDRC-2.

7) MDRC-6: Radio Applications have static Resource requirements, on-device compilation of Software Radio Components

This class corresponds to MDRC-3 with the difference that all or part of the software radio components are provided in the Radio Application Package as platform-independent source code or platform-independent IR, which is compiled on the reconfigurable MD itself. As in the case of MDRC-5, it particularly means that the reconfigurable MD should include a proper compiler in order to convert the source code or IR of the software radio components into an executable code that runs on a given modem chip of a reconfigurable MD. As in the case of MDRC-5, it is assumed that the methods of radio programming and the tools to support this category have become sufficiently standardized so that third-party vendors may create Radio Applications and activate them to different platforms with relative ease. The formal description of the Resource management is the same as for MDRC-3.

8) MDRC-7: Radio Applications have dynamic Resource requirements, on-device compilation of Software Radio Components

This class corresponds to MDRC-4 with the difference that all or part of the software radio components are provided in the Radio Application Package as platform-independent source code or platform-independent IR, which is compiled on the reconfigurable MD itself. As in the case of MDRC-5 or MDRC-6, it particularly means that the reconfigurable MD should include a proper compiler in order to convert the source code or IR of the software radio components into an executable code that runs on a given modem chip of a reconfigurable MD. As in the case of MDRC-5 or MDRC-6, it is assumed that the methods of radio programming and the tools to support this category have become sufficiently standardized so that third-party vendors may create Radio Applications and activate them to different platforms with relative ease. The formal description of the Resource management is the same as for MDRC-4.

The definition of MDRCs described above can be summarized as shown in Table 1.

Table 1: Summary of MDRCs

<table>
<tr><th></th><th>Multi-radio system</th><th>Resource Share (among Radio Applications)</th><th>Resource Manager</th><th>Multi-tasking</th><th>Resource Measurement</th><th>Resource Allocation</th></tr>
<tr><td>MDRC-0</td><td>No</td><td>No</td><td>No</td><td>No</td><td>Design-time</td><td>Design-time</td></tr>
<tr><td>MDRC-1</td><td>Yes</td><td>No</td><td>No</td><td>No</td><td>Design-time</td><td>Design-time</td></tr>
<tr><td rowspan="2">MDRC-2
MDRC-5</td><td rowspan="2">Yes</td><td rowspan="2">No
(note 1)</td><td rowspan="2">Yes
(note 2)</td><td rowspan="2">Yes
(note 3)</td><td>Design-time</td><td>Design-time</td></tr>
<tr><td>Design-time
/Install-time</td><td>Design-time
/Install-time</td></tr>
<tr><td rowspan="2">MDRC-3
MDRC-6</td><td rowspan="2">Yes</td><td rowspan="2">Yes</td><td rowspan="2">Yes</td><td rowspan="2">Yes</td><td>Design-time</td><td rowspan="2">Run-time</td></tr>
<tr><td>Design-time
/Install-time</td></tr>
<tr><td rowspan="2">MDRC-4
MDRC-7</td><td rowspan="2">Yes</td><td rowspan="2">Yes</td><td rowspan="2">Yes</td><td rowspan="2">Yes</td><td>Design-time</td><td rowspan="2">Run-time</td></tr>
<tr><td>Design-time
/Install-time</td></tr>
<tr><td colspan="7">NOTE 1: Resource share can exist among Radio Access Technologies (RATs) in a given Radio Application.
NOTE 2: This is for a fixed Resource allocation only. Resource management and Resource allocation among RATs (in a single RA) are pre-determined in a static manner by Radio Application provider.
NOTE 3: Multi-tasking in this case is for multiple RATs within a single Radio Application.</td></tr>
</table>

Note that radio conformance tests are mandatory for MDRC-1 to MDRC-7 in order to ensure that the joint operation of (dynamically) reconfigured base-bands and RF front-ends are in compliance with the relevant conformance requirements before the device is introduced into the market.

The requirements described in the present document are based on the MDRCs above defined. As it can be noted in clause 6, some requirements are independent from the class of device while others apply only to well defined classes. Therefore a reconfigurable MD will follow only those requirements related to the MDRC it belongs to.

6 Functional requirements

6.1 Requirements on RAT link support and management

6.1.1 R-FUNC-RAT–01 Function for MDRC-1 to MDRC-7

A reconfigurable MD should support parallel connections to more than one Radio Access Technology.

6.1.2 R-FUNC-RAT–02 Function for MDRC-1 to MDRC-7

If a reconfigurable MD allows parallel connections to RATs, (in alignment to R-FUNC-RAT-01), in-device coexistence functionalities shall be implemented.

6.1.3 R-FUNC-RAT–03 Function for MDRC-1 to MDRC-7

If a reconfigurable MD allows parallel connections to RATs (in alignment to R-FUNC-RAT-01), seamless handover of data streams from one RAT to another RAT should be implemented.

NOTE: Seamless handover is only between RATs used for the same service.

Explanation: A seamless handover does not create any interruption of an ongoing service.

6.1.4 R-FUNC-RAT–04 Function for MDRC-1 to MDRC-7

If policies are applied to a reconfigurable MD, the link selection functionality in the reconfigurable MD shall meet the related conditions.

Explanation: It is possible to provide policies to reconfigurable MDs. These policies introduce link selection constraints to be met by the link selection decision making functionality in the concerned reconfigurable MD. For example, the policies may be used for enforcing Network Operator preferences, User preferences, etc.

6.1.5 R-FUNC-RAT–05 Function for MDRC-1 to MDRC-7

If a reconfigurable MD allows parallel connections to RATs (in alignment to R-FUNC-RAT–01), various independent data flows should be maintained simultaneously.

6.1.6 R-FUNC-RAT–06 Function for MDRC-1 to MDRC-7

If a reconfigurable MD allows parallel connections to RATs (in alignment to R-FUNC-RAT–01), Link Adaptation techniques across multiple RATs should be implemented.

NOTE: Link adaptation techniques include network coding, air interface selection, offloading, etc.

6.2 Radio application requirements

6.2.0 General

The following requirements are based on the Use Cases described in ETSI TR 102 944 [i.2].

6.2.1 R-FUNC-RA-01 Radio Applications Support for MDRC-1 to MDRC-7

Reconfigurable MDs shall support the execution of Radio Applications.

6.2.2 R-FUNC-RA-02 Composition for MDRC-1 to MDRC-7

Radio Applications shall be composed of SFBs and/or UDFBs.

Explanation: The resulting Radio Applications will be composed by combining SFBs and/or UDFBs. They will be used, for example, for baseband signal processing and real-time Media Access Control (MAC) data processing.

6.2.3 R-FUNC-RA-03 Concurrency for MDRC-1 to MDRC-7

Reconfigurable MDs should support concurrent execution of Radio Applications.

6.2.4 R-FUNC-RA-04 Data for MDRC-1 to MDRC-7

Radio Applications should support the function of transferring receive (Rx)/transmit (Tx) data to/from the networking stack.

6.2.5 R-FUNC-RA-05 Context Information for MDRC-1 to MDRC-7

Radio Applications should support the function of delivering context information.

Explanation: Access to context information metrics is provided. As an example, the metrics can be related to:

- Signal to Interference-plus-Noise Ratio (SINR)
- Received Signal Strength Indication (RSSI)
- Packet Error Rate (PER)
- Bit Error Rate (BER)
- Power consumption of selected Base-Band modules
- MIMO algorithm configuration indicator
- Precoding Matrix Indicator (PMI)
- Rank Indicator (RI)
- Data rate indicator
- Channel Coefficients Information

6.2.6 R-FUNC-RA-06 Pipelining for MDRC-2 to MDRC-7

Radio Applications shall be applicable to fixed pipeline, programmable pipeline, and hybrid pipeline architectures.

Explanation: The pipeline of functional blocks is determined by the contents of metadata included in the Radio Application Package. Modem chip manufacturer selects one of the 3 pipeline structures for their modem chip processor. When the modem chip processor is configured according to the contents of metadata, the configuration of modem chip processor is performed according to the pipeline structure adopted in a given modem chip.

6.3 Radio Application functional block requirements

6.3.1 R-FUNC-FB-01 Implementation for MDRC-2 to MDRC-7

Each instance of a functional block shall be implemented with a corresponding program code characterized by the implementation properties and shall be accessed by a corresponding interface.

Explanation: The typical structure of the functional block is shown in Figure 3. The corresponding interface is represented as "*Name(Input, Output)*" and the functionality of each functional block is implemented with the corresponding program code attributed by the properties.

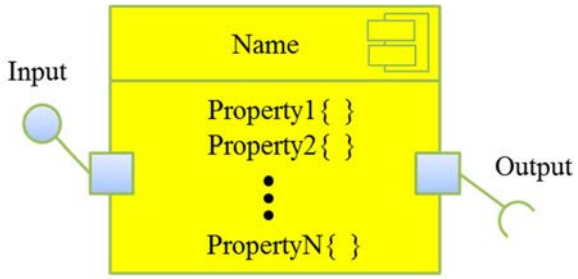


Figure 3: Functional block defined as a baseband signal processing component

6.3.2 R-FUNC-FB-02 Execution for MDRC-2 to MDRC-7

Each instance of a functional block shall be executed only by calling the corresponding interface.

Explanation: The execution of an instance of a functional block requires an external trigger, for example in order to avoid recursion.

6.3.3 R-FUNC-FB-03 Side Effects for MDRC-2 to MDRC-7

The internal state of an instance of each functional block shall not be shared with any other instances of functional blocks.

Explanation: The behaviour of an instance of a functional block depends only on its inputs and not on the state of another instance of a functional block.

6.3.4 R-FUNC-FB-04 Shared Data for MDRC-2 to MDRC-7

The reconfigurable MD should support data sharing with other instances of functional blocks.

Explanation: Input and/or output of each functional block can be shared with other instances of functional block(s).

6.3.5 R-FUNC-FB-05 Concurrency for MDRC-2 to MDRC-7

The reconfigurable MD should support the concurrent execution of instances of functional blocks.

6.3.6 R-FUNC-FB-06 Extendibility for MDRC-2 to MDRC-7

A library extension shall be supported.

Explanation: A Radio Application consists of UDFBs as well as SFBs. Meanwhile, the Radio Library of a reconfigurable MD consists of SFBs only and not of UDFBs. Therefore, the reconfigurable MD needs a normative library extension to support UDFBs. In other words, the reconfigurable MD needs to support a normative procedure for a normative library extension.
In case of MDRC-5 to MDRC-7, the library extension is performed on the reconfigurable MD.

6.4 Mobile device reconfiguration requirements

6.4.1 R-FUNC-MDR-01 Platform-specific Executable Code for MDRC-2, MDRC-3 or MDRC-4

The configuration of a reconfigurable MD compliant to MDRC-2, MDRC-3 or MDRC-4 shall be realized with a Radio Application Package of which the user defined functional blocks, if any, are provided in platform-specific executable code.

Explanation: A Radio Application Package provided with the platform-specific executable code is designed specifically for each kind of target hardware platform. Each platform-specific executable code is generated using a specific compiler for the corresponding target hardware platform during design time. Then, the reconfigurable MD configuration is performed through downloading of the compiled code into the reconfigurable MD and its installation.

6.4.2 R-FUNC-MDR-02 Platform-independent Source Code or IR for MDRC-5, MDRC-6 or MDRC-7

The configuration of a reconfigurable MD compliant to MDRC-5, MDRC-6 or MDRC-7 shall be realized with a Radio Application package of which the user defined functional blocks, if any, are provided either in a platform-independent source code or an Intermediate Representation (IR).

Explanation for Source Code case:

A Radio Application package can adopt user defined functional blocks in a form of platform-independent source code with a proper encryption. Since a Radio Application package is provided in platform-independent source code, the user defined functional block code is compiled to generate corresponding executable code using a compiler for the specific reconfigurable MD hardware platform. Note that standard functional blocks are based on a reference design given by the Radio Library that is given in platform specific executable code.

Explanation for IR case:

A Radio Application package adopting a platform-independent IR is targeted to different kinds of modem chips. Since a Radio Application package is provided in a platform-independent IR (i.e. non-executable code), the platform-independent IR is always translated into executable code using a back-end compiler for the specific reconfigurable MD hardware platform. Note that standard functional blocks are based on a reference design given by the Radio Library that is given in platform specific executable code.

6.4.3 R-FUNC-MDR-03 Radio Configuration of Platform MDRC-1 to MDRC-7

The radio configuration of a reconfigurable MD shall be realized with the activation of Radio Applications (RA) and, if necessary, changing parameters of the activated RAs.

6.4.4 R-FUNC-MDR-04 Radio Programming for MDRC-1 to MDRC-7

The reconfigurable MD shall provide a suitable interface which conveys structural and behavioural information of RAs for the reconfigurable MD reconfiguration.

Explanation: The interface enables the provision of reconfiguration code onto the reconfigurable MD. The structural information defines a set of functional blocks or RA computational operators, data and communications among them. The behavioural information defines execution rules and operator interactions.

6.4.5 R-FUNC-MDR-05 Dynamic Execution for MDRC-4, and MDRC-7

The reconfigurable MD shall support dynamic execution of functional blocks.

Explanation: In case of dynamic Resource sharing, the Resource allocation is performed in run time.

6.4.6 R-FUNC-MDR-06 Independency on Memory Model for MDRC-1 to MDRC-7

Different radio platform memory models shall be supported. Those shall include the shared memory and the message passing memory models.

6.4.7 R-FUNC-MDR-07 Code for MDRC-2 to MDRC-7

The interface related to R-FUNC-MDR-04 shall support the provision of executable code (for MDRC-2 to MDRC-4) and IR and/or Source Code (for MDRC-5 to MDRC-7).

6.4.8 R-FUNC-MDR-08 IR Format for MDRC-5 to MDRC-7

IR shall be defined in a format suitable for human reading/writing and automated processing.

6.4.9 R-FUNC-MDR-09 Timing Constraints for MDRC-1 to MDRC-7

The interface related to R-FUNC-MDR-04 shall support timing constraints for radio processing.

Explanation: This requirement relates to real time execution. The execution of concerned functional blocks will be in accordance to the provided timing constraints.

6.4.10 R-FUNC-MDR-10 Platform Independency for MDRC-5 to MDRC-7

The reconfigurable MD architecture shall provide suitable interfaces in order to ensure platform independency.

6.4.11 R-FUNC-MDR-11 Radio Application for MDRC-5 to MDRC-7

The Software representation (i.e. IR or Source Code) of an RA shall be independent of the target hardware platform.

Explanation: The independence of the RA from the target hardware platform indicates that the RA can be provided to multiple platforms supporting the interfaces related to R-FUNC-MDR-10.

6.4.12 R-FUNC-MDR-12 Function Granularity for MDRC-1 to MDRC-7

Functional blocks may have different granularity.

Explanation: The functional blocks might be for example arithmetic operations or functions like FFT or even whole Radio Applications like WiFi, LTE, etc.

6.4.13 R-FUNC-MDR-13 Radio Virtual Machine for MDRC-2 to MDRC-7

Radio Application(s), SFB(s) or UDFB(s) shall be executed on a suitably configured Radio Virtual Machine, including the application of a suitable protection class.

Explanation: A RVM execution environment should be provided in such a way that a proper RVM (protection) class can be selected by each vendor. The RVM approach is required for platform dependent and/or independent 3rd party code, since a manufacturer will require that 3rd party code is executed in a controlled environment. The 3rd party code corresponds to a Radio Application, SFB(s) or UDFB(s).

6.4.14 R-FUNC-MDR-14 RadioVirtual Machine Structure for MDRC-2 to MDRC-7

A Radio Virtual Machine may consist of several smaller Radio Virtual Machines.

Explanation: There may be an RVM hierarchy, i.e. one or several RVMs might be a part of the bigger RVM. A smaller RVM is connected to external RVM ports of the bigger RVM.

6.4.15 R-FUNC-MDR-15 Selection of Radio Virtual Machine Protection Class for MDRC-2 to MDRC-7

A Radio Application shall select a suitable Radio Virtual Machine Protection Class.

Explanation: Radio Virtual Machine (protection) classes are introduced in order to find a trade-off between (re-) certification effort and base-band code development flexibility.

At one extreme of RVM class, a high-level RVM class corresponds to full reconfigurability of the low-level parameters of an RVM, and accordingly necessitates a relatively more extensive certification testing process after the RVM has been reconfigured. At the other extreme of RVM class, a low-level RVM class corresponds to a limited reconfigurability of the low-level parameters of an RVM. As the reconfigurability of the low-level parameters of this particular class of RVM is limited, a relatively less extensive certification testing process is necessitated after the RVM has been reconfigured. Moreover, an RVM can have different RVM classes associated with different components of the RVM that relates to the reconfigurability of the low-level parameters of the respective components of the RVM.

Reconfiguration of an RVM of the highest-level RVM class may necessitate that the overall certification testing process focuses on the certification of each reconfigured software components of the RVM. In such a situation, each respective reconfigured software component may need to be separately certified before one or more sets of reconfigured software components are certified together. For example, a reconfigured RVM software component "A" (e.g. WiFi) may need to be separately certified from a reconfigured and certified RVM software component "B" (e.g. LTE). The certification process may then be such that the joint operation of separately certified reconfigured RVM software components "A" and "B" may then take place jointly.

At the other extreme of RVM classes, the lowest-level RVM class corresponds to a restricted reconfigurability of the low-level parameters of an RVM. For such a restricted level of reconfigurability, a developer of Radio Applications would only have limited access to the low-level parameters of an RVM. For example, the lowest-level RVM class would permit a Radio Application developer to have access to only the low-level parameters of the receive chain of an RVM. Accordingly, the lowest-level of RVM class would not need to utilize a corresponding detailed and thorough certification testing process because, for example, a radio platform operating a malfunctioning reconfigured RVM would not interfere with other radio platforms. Thus, level of certification testing for the lowest RVM class would be less extensive certification testing process than that used for the highest RVM class.

One or more medium- or intermediate-level RVM classes may also be established between the two extreme RVM classes that correspond to intermediate levels of reconfigurability of the low-level parameters of an RVM. An intermediate-level RVM class, for example, would allow more flexibility for reconfiguring low-level parameters of an RVM than the lowest-level RVM class, but would not permit the degree of reconfigurability that would be associated with the highest-level RVM class. Depending on the level of reconfigurability to the low-level parameters of an RVM, an intermediate-level RVM class may necessitate a certification testing process for a compiled reconfigured RVM and underlying hardware that is more extensive than that corresponding to the lowest-level RVM class, but less extensive than that corresponding to the highest-level RVM class. For example, a certification based on the intermediate-level RVM software component might be obtained by contacting an authorized notified body and providing only a serial number for the RVM software component and an identification of the target device type on which the compiled reconfigured RVM would operate. In another example, there could be no requirement for a joint certification based on an RVM software component for a simultaneous operation with other RVM software components. That is, a certificate based on an RVM software component "A" (e.g. WiFi) and a separate certificate based on another RVM software component "B" (e.g. LTE) would allow for a simultaneous operation of reconfigured components "A" (e.g. WiFi) and "B" (e.g. LTE).

Another exemplary situation that may necessitate a relatively less extensive certification testing process would be a Radio Application developer that only reconfigures non-transmission-related low-level parameters, for example, low-level parameters relating to a data interleaver and/or a channel coder in the transmit/receive (TX/RX) chain of an RVM that otherwise has been defined to be of the highest-level RVM class. As nothing related to the spectral shaping of a transmitted signal is reconfigured by the reconfiguration of the data interleaver and/or channel coder, a relatively less extensive certification testing process could be used. Another exemplary situation that may necessitate a less extensive certification testing process would be a reconfiguration that involves changes targeting predefined frequency bands and/or bandwidths. In still other exemplary situations, there may be reconfigurations for which a certification testing process may not be necessary.

6.5 Radio Frequency(RF) transceiver requirements

6.5.0 General

The following requirements are related to the RF transceiver which allows for the selection of RF configuration parameters. However, it does not limit the manufacturer in the choice of supported features in a given platform.

6.5.1 R-FUNC-RFT-01 RF Configuration for MDRC-1 to MDRC-7

The reconfigurable MD shall provide a suitable interface for RF transceiver configuration.

Explanation: The interface will enable the exchange of control and data information between the Radio Applications and the RF transceiver.

6.5.2 R-FUNC-RFT-02 Extendibility for multiple-antenna system for MDRC-1 to MDRC-7

If a reconfigurable MD supports multiple antenna operation, a Radio Application shall be able to select a suitable number of antenna inputs/outputs.

Explanation: The 3rd party developers can choose among multiple antenna technologies such as, for example, SU-MIMO, MU-MIMO, massive MIMO, etc. The extendibility of physical antennas relates to the number of RF input/output signals from/to the RF front-end(s) to to/from the antennas.

6.5.3 R-FUNC-RFT-03 Capability of multiple frequency bands for MDRC-1 to MDRC-7

The reconfigurable MD shall support multiple Radio Applications using distinct frequency bands.

6.5.4 R-FUNC-RFT-04 Reconfigurability of RF Transceiver for MDRC-1 to MDRC-7

An RF transceiver shall manage input/output signals from/to one or several Radio Applications.

Explanation: Several Radio Applications which are simultaneously in active state may be served by one or multiple RF transceivers.

6.5.5 R-FUNC-RFT-05 Interoperability of radio resources for MDRC-2 to MDRC-7

Sharing of radio resources among multiple Radio Applications shall be supported.

6.5.6 R-FUNC-RFT-06 Testability of radio equipment for MDRC-1 to MDRC-7

A test mode in which the transmitter chain is connected to the receiver chain in loop-back mode shall be supported.

Explanation: The test mode provides test capability of the RF path without actual radio waves emission.

6.5.7 R-FUNC-RFT-07 Unified representation of control information for MDRC-1 to MDRC-7

The interface related to R-FUNC-RFT-01 shall support a unified representation of control information passed to/from the RF front-end.

Explanation: Control information passed to/from the RF front-end is represented in a unified format suitable for RF front-end handling.

6.5.8 R-FUNC-RFT-08 Unified representation of data payload for MDRC-1 to MDRC-7

The interface related to R-FUNC-RFT-01 shall support a unified representation of data payload passed to/from the RF front-end.

Explanation: Data payload passed to/from the RF front-end is represented in a unified format suitable for RF front-end data handling.

6.5.9 R-FUNC-RFT-09 Selection of RF Protection Class for MDRC-1 to MDRC-7

The interface related to R-FUNC-RFT-01 shall support a suitable selection of an RF Protection Class.

Explanation: RF protection classes are introduced in order to find a trade-off between (re-)certification effort and RF front-end flexibility. In case of a low RF Protection Class, the RF front-end provides a high level of flexibility (e.g. in terms of band selection, bandwidth selection, out-of-band radiation, etc.) which typically requires a thorough and complex (re-)certification of the concerned Radio Applications. In case of a high RF Protection Class, on the other hand, the RF front-end typically introduces protection mechanisms (e.g. filters for limiting out-of-band radiation, etc.) which limit the flexibility of the RF front-end but allow for a lighter (re-)certification of corresponding Radio Applications.

6.6 Security requirements

6.6.0 General

The following requirements are related to End-to-End Security which allows to ensure that only approved Radio Applications may be downloaded to and executed on a target Mobile Device.

6.6.1 R-FUNC-SEC-01 REConfPol-RAP-Security

Cryptographic functions shall be provided to support the digital signature strategy providing confidentiality, integrity, and authenticity of RE Configuration Policy and RAP(s), and the non-repudiation strategy.

6.6.2 R-FUNC-SEC-02 Administration-Security

A security function in the Communication Service Layer shall ensure that only approved RAP(s) are processed and executed on a target platform.

Explanation: Typically, an *Administrator Security Function* is added to the *Administrator* entity in the *Communication Service Layer*.

6.6.3 R-FUNC-SEC-03 Secure Management

Security functions shall be provided to allow remote parties to securely attest to the status of the Mobile Device, configure the Mobile Device, and manage it over extended period of time.

Explanation: Typically, an *Attesting Entity* is present on the device (possibly in a distributed fashion), that can act as a root of trust for reporting. The *Administrator* entity in the *Communication Service Layer* is able to receive, validate, and act on configuration commands from authorized remote parties.

The flexibility introduced by radio reconfiguration will greatly extend the lifetime of the device. It shall be possible to bind the device to a new configuration authority when necessary. Typically, an *RRS Configuration Manager* is added to the *Administrator* entity in the *Communication Service Layer* for the purpose of keeping track of active configuration authorities.

6.6.4 R-FUNC-SEC-04 Root of Trust

A root of trust shall provide security services to the reconfigurable platform.

Explanation: Components are required in order to provide a high level of security assurance throughout the lifecycle of the Mobile Device.

History

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